Problem: Properties, Rank, and Spectrum of a Matrix

Problem Statement

Let $\phi \in \mathbb{R}$. Consider the matrix

$$A = \begin{pmatrix} 0 & e^{-i\phi} \\ e^{i\phi} & 0 \end{pmatrix}.$$

(i) Is the matrix A Hermitian? Is it unitary?

(ii) Find the rank of the matrix.

(iii) Find the eigenvalues and eigenvectors of A.

(iv) Let I_2 be the 2×2 identity matrix. Find the eigenvalues of $A \otimes I_2$.

(i) Hermitian and Unitary Properties

Hermitian Test

A matrix is Hermitian if

$$A^\dagger = A,$$

where $A^\dagger$ denotes the conjugate transpose.

Compute $A^\dagger$:

$$A^\dagger = \begin{pmatrix} 0 & e^{-i\phi} \\ e^{i\phi} & 0 \end{pmatrix} = A.$$

$$\boxed{A \text{ is Hermitian for all } \phi \in \mathbb{R}.}$$

Unitary Test

A matrix is unitary if

$$A^\dagger A = I.$$

Since $A^\dagger = A$, compute:

$$A^2 = \begin{pmatrix} 0 & e^{-i\phi} \\ e^{i\phi} & 0 \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I.$$

$$\boxed{A \text{ is unitary for all } \phi \in \mathbb{R}.}$$

Insight

$$A = \cos\phi\,\sigma_x + \sin\phi\,\sigma_y,$$

so A represents a Pauli-type observable: Hermitian (observable) and unitary (norm-preserving).

(ii) Rank of the Matrix

The determinant is:

$$\det(A) = -e^{-i\phi}e^{i\phi} = -1 \neq 0.$$

Since the determinant is nonzero,

$$\boxed{\operatorname{rank}(A) = 2.}$$

(iii) Eigenvalues and Eigenvectors

Eigenvalues

Solve the characteristic equation:

$$\det(A - \lambda I) = \begin{vmatrix} -\lambda & e^{-i\phi} \\ e^{i\phi} & -\lambda \end{vmatrix} = \lambda^2 - 1 = 0.$$

$$\boxed{\lambda = \pm 1.}$$

Eigenvectors

For $\lambda = +1$: Solve $(A - I)x = 0$:

$$\begin{pmatrix} -1 & e^{-i\phi} \\ e^{i\phi} & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0.$$

A solution is:

$$x_+ = \begin{pmatrix} e^{-i\phi} \\ 1 \end{pmatrix}.$$

For $\lambda = -1$: Solve $(A + I)x = 0$:

$$\begin{pmatrix} 1 & e^{-i\phi} \\ e^{i\phi} & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0.$$

A solution is:

$$x_- = \begin{pmatrix} -\,e^{-i\phi} \\ 1 \end{pmatrix}.$$

(iv) Eigenvalues of $A \otimes I_2$

Key Property

If λ is an eigenvalue of A and μ is an eigenvalue of I_2, then

$$\lambda\mu$$

is an eigenvalue of $A \otimes I_2$.

Application

Eigenvalues of A: ± 1
Eigenvalues of I_2: $1, 1$

Thus the eigenvalues of $A \otimes I_2$ are:

$$\boxed{+1,\ +1,\ -1,\ -1.}$$

Insight

Tensoring with the identity *duplicates* the spectrum without changing its values.

Final Summary

Property	**Result**
Hermitian	Yes (for all $\phi \in \mathbb{R}$)
Unitary	Yes (for all $\phi \in \mathbb{R}$)
Rank	2
Eigenvalues of A	± 1
Eigenvalues of $A \otimes I_2$	± 1 (each twice)

Key Conceptual Insight

$$\boxed{\text{Hermitian} + \text{Unitary} \ \Rightarrow \text{Eigenvalues lie on } \{\pm 1\}}$$

Such matrices naturally arise as quantum observables and symmetry operators.